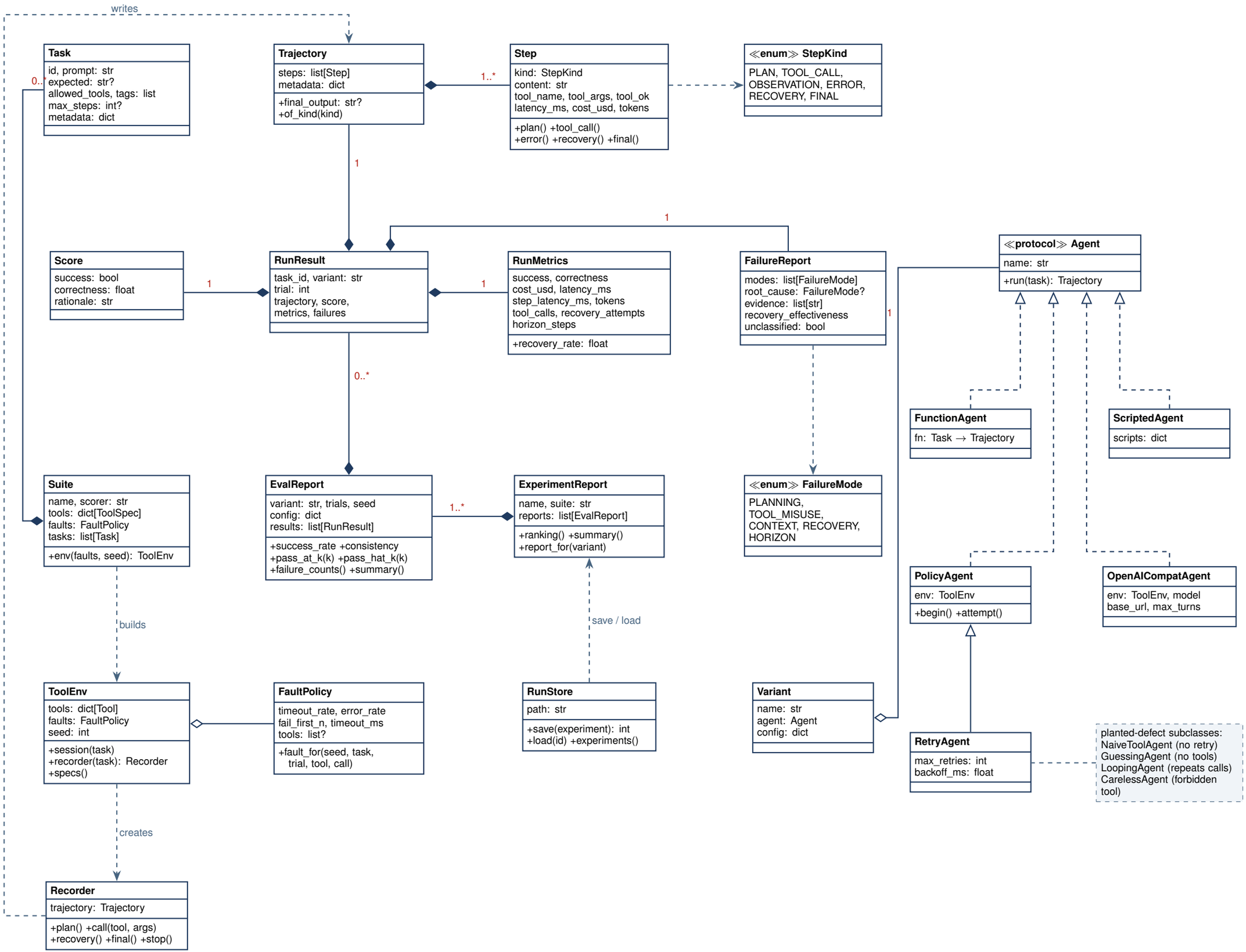
writes
Task
id, prompt: str
expected: str?
allowed_tools, tags: list
max_steps: int?
metadata: dict
0..
Trajectory
steps: list[Step]
metadata: dict
+final_output: str?
+of_kind(kind)
1..*
Step
kind: StepKind
content: str
tool_name, tool_args, tool_ok
latency_ms, cost_usd, tokens
+plan() +tool_call()
+error() +recovery() +final()
≪enum≫ StepKind
PLAN, TOOL_CALL,
OBSERVATION, ERROR,
RECOVERY, FINAL
1
1
Score
success: bool
correctness: float
rationale: str
1
RunResult
task_id, variant: str
trial: int
trajectory, score,
metrics, failures
1
RunMetrics
success, correctness
cost_usd, latency_ms
step_latency_ms, tokens
tool_calls, recovery_attempts
horizon_steps
+recovery_rate: float
FailureReport
modes: list[FailureMode]
root_cause: FailureMode?
evidence: list[str]
recovery_effectiveness
unclassified: bool
1
0..*
≪protocol≫ Agent
name: str
+run(task): Trajectory
FunctionAgent
fn: Task → Trajectory
ScriptedAgent
scripts: dict
Suite
name, scorer: str
tools: dict[ToolSpec]
faults: FaultPolicy
tasks: list[Task]
+env(faults, seed): ToolEnv
EvalReport
variant: str, trials, seed
config: dict
results: list[RunResult]
+success_rate +consistency
+pass_at_k(k) +pass_hat_k(k)
+failure_counts() +summary()
1..*
ExperimentReport
name, suite: str
reports: list[EvalReport]
+ranking() +summary()
+report_for(variant)
≪enum≫ FailureMode
PLANNING,
TOOL_MISUSE,
CONTEXT, RECOVERY,
HORIZON
builds
save / load
PolicyAgent
env: ToolEnv
+begin() +attempt()
OpenAICompatAgent
env: ToolEnv, model
base_url, max_turns
ToolEnv
tools: dict[Tool]
faults: FaultPolicy
seed: int
+session(task)
+recorder(task): Recorder
+specs()
FaultPolicy
timeout_rate, error_rate
fail_first_n, timeout_ms
tools: list?
+fault_for(seed, task,
trial, tool, call)
RunStore
path: str
+save(experiment): int
+load(id) +experiments()
Variant
name: str
agent: Agent
config: dict
RetryAgent
max_retries: int
backoff_ms: float
planted-defect subclasses:
NaiveToolAgent (no retry)
GuessingAgent (no tools)
LoopingAgent (repeats calls)
CarelessAgent (forbidden
tool)
creates
Recorder
trajectory: Trajectory
+plan() +call(tool, args)
+recovery() +final() +stop()